
AMR ASHOUR, PhD

Hydrogeologist | Groundwater Specialist | Hydrologist | Flood Protection Specialist

Location: Nasr City, Cairo, Egypt

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Nationality: Egyptian | Date of Birth: 11 January 1986

PROFESSIONAL OBJECTIVE

Highly experienced **Civil Engineer** specializing in **Hydrology, Hydrogeology, and Groundwater Engineering**, with extensive professional experience in **national and international projects** across **Egypt, Saudi Arabia, UAE, Iraq, Qatar, Rwanda, Pakistan, and Tanzania**. Demonstrated expertise in **hydrological and hydrogeological studies, groundwater modeling, flood protection systems, stormwater drainage networks, groundwater control and dewatering, hydraulic calculations, and irrigation works**. Strong academic and professional background in **environmental engineering, karst aquifers, coastal groundwater systems, seawater intrusion, groundwater-surface water interaction, and integrated water resources management**, with proven ability to deliver **authority-approved technical reports, engineering designs, and sustainable water solutions** fully aligned with international standards and **ATS requirements**.

CORE TECHNICAL COMPETENCIES

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| <ul style="list-style-type: none">• Hydrological Studies• Stormwater Drainage Systems• Hydraulic Calculations• Hydraulic Structures Design• Integrated Water Resources Management• Technical Reports Preparation• Early Warning Systems for Flash Floods• Flood Protection & Flash Flood Assessment• Project Coordination & Site Supervision | <ul style="list-style-type: none">• Hydrogeological Investigations• Groundwater Modeling• Environmental Engineering• Karst Aquifer Analysis• Coastal Groundwater Systems• Seawater Intrusion Analysis• Groundwater-Surface Water Interaction• Groundwater Control & Dewatering Systems• Groundwater-Surface Water Interaction |
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PROFESSIONAL EXPERIENCE

Lecturer in Civil Engineering (Hydrology, Water Resources & Groundwater)

Faculty of Engineering, Al-Azhar University – Cairo, Egypt

Key Achievements & Responsibilities

- Delivered academic instruction in **Hydrology, Hydrogeology, Hydraulic Engineering, and Water Resources Engineering**.
- Developed lecture materials, course outlines, and technical notes aligned with modern engineering standards.
- Supervised laboratory experiments and practical training to enhance applied engineering competencies.
- Provided technical and academic supervision for undergraduate graduation projects.
- Contributed to applied research in **groundwater modeling and hydrological analysis**.
- Participated in academic committees, examinations, and student evaluation processes.
- Supported scientific research dissemination and peer academic collaboration.

Key Projects / Academic Activities

- Graduation Project Supervision – Hydrology & Water Resources
 - Applied Research in Groundwater Flow Modeling
 - Academic Laboratories Development & Practical Training
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Project Manager | Groundwater Lowering Systems

ALBWARDI ENGINEERING CONSULTING– Riyadh, KSA

Aug 2025- Jan 2026 (Full Time)

Key Achievements & Responsibilities

- Led and supported critical project stages from **draft design** through **final design** and **tender preparation** for **groundwater lowering and control systems**.
- Delivered engineering support for **groundwater control**, initial **pump station design** and system design review.

- Prepared **technical documentation** including design calculations, reports, specifications, and tender-related deliverables.
- Contribute to the **value engineer workshops** to select the optimum solution for the groundwater control system
- Reviewed **environmental studies** to ensure design outputs met technical requirements, project standards, and environmental compliance obligations.
- **Coordinated with clients**, multidisciplinary teams, and project stakeholders to **achieve timely and accurate design completion**.
- Supported **quality control**, **value engineering**, and **commercial review** activities during final design and tender stages.
- Contributed to **pricing / cost estimation** and demonstrated strong **commercial awareness** throughout project development.

Key Projects

- Groundwater Control and lowering Studies – (Yanbu1, Yanbu2 and Ar Rayis), **Madinah, KSA**.
- Groundwater Control and lowering Studies – (Al Qunfudhah1, Al Qunfudhah2 and Al Qunfudhah3), **Jeddah, KSA**.
- Groundwater Control and lowering Studies – (Jazan1 and Jazan2), **Jazan, KSA**.
- Groundwater Control and lowering Studies – (Umluj and Haql), **Tabuk, KSA**.
- Groundwater Control and lowering Studies – (Al Birk), **Aseer, KSA**.

Senior Hydrologist / Hydrogeologist

Khatib & Alami Consultant Company – Cairo Branch, Egypt

2019 – 2020 (Full Time) | 2022 – Present (Per Task)

Key Achievements & Responsibilities

- Led comprehensive **hydrological and hydrogeological studies** for large-scale infrastructure and urban developments.
- Delivered integrated **flood protection and flash flood mitigation studies**, including watershed analysis and rainfall-runoff modeling.
- Developed and calibrated **groundwater numerical models** to simulate aquifer behavior under development scenarios.
- Executed advanced **groundwater mounding, seepage analysis, groundwater lowering, and control studies**.
- Prepared detailed **technical reports, engineering calculations, drawings, specifications, and method statements**.
- Reviewed designs for compliance with **local regulations and international standards**.
- Coordinated with multidisciplinary teams and international stakeholders.
- Supported projects during **authority review, value engineering, and construction stages**.

Key Projects

- Flood Protection Projects – Al Sahn, El Derbin, Al Farain, Al AEZ at Assir, KSA
- Storm Drainage Network Assessment – Medan Racecourse, Dubai, UAE
- Storm Drainage Network Assessment – El Khawaneej, Abu Dhabi, UAE
- Groundwater Mounding & Lowering Studies – Basra City, Iraq
- Groundwater Control Studies – El Wathba North & South, UAE
- Value Engineering Study – Etihad Railway Project (139 km), UAE-KSA
- Groundwater Control Study – Asdaf Obhor Residential Project, Jeddah, KSA

Senior Hydrology, Stormwater Drainage & Groundwater Engineer | Office Manager

Vision Dimensions Consultant Office – Egypt Branch

August 2019 – May 2023

Key Achievements & Responsibilities

- Managed and delivered **flood protection and stormwater drainage projects** for major urban developments.
- Planned, designed, and reviewed **stormwater drainage networks**, including hydraulic verification.
- Led **groundwater lowering and groundwater control studies** for construction and long-term development.
- Supervised preparation and submission of **engineering reports and authority approval documents**.
- Delivered professional technical lectures on **groundwater engineering and environmental impacts**.
- Coordinated with clients, municipalities, and regulatory authorities.

- Managed engineering teams, task allocation, and quality control processes.

Key Projects

- Flood Protection Projects – Nowaisr, El Shimisy, Water Treatment Project, Jeddah, KSA
- Flood Protection Project – Wadi Labn, Riyadh, KSA
- Stormwater Drainage Networks – Private schemes (654–677-679-680-681-682-683), Jeddah, KSA
- Groundwater Lowering Study – Al Habib Residential Community, Jeddah, KSA
- Groundwater Control Study (Phase 4) – Al Rayan District, Jeddah, KSA
- City-Wide Stormwater and Groundwater Network Survey – Jeddah, KSA
- Professional Lectures – Jeddah Municipality, KSA
- Dewatering Systems Design – More than 20 project – Jeddah, KSA

Hydrological and Stormwater Drainage Design Engineer – Freelance

Arkon Engineering Consultants

2024 - 2025

Key Achievements & Responsibilities

- Managed and delivered **flood protection and stormwater drainage projects** for major urban developments.
- Planned, designed, and reviewed **stormwater drainage networks**, including hydraulic verification.

Key Projects

- Flood Protection Project – Jawhara Mall, Riyadh, KSA
- Flood Protection Project – Jabal Mecca Building, Mecca, KSA
- Flood Protection Project – Al Qwizain Tomb, Jeddah, KSA
- Stormwater Drainage Networks – STC Parking and Private schemes (322, 389 & 718), Jeddah, KSA

Hydrogeologist

CIVILSTRICT Engineering Consultant Office

January 2018 – December 2018

Key Achievements & Responsibilities

- Executed **groundwater control, extraction, and dewatering studies** for infrastructure and residential projects.
- Designed and reviewed **temporary and permanent dewatering systems**.
- Conducted **pumping tests, slug tests, and aquifer performance analysis**.
- Prepared **hydrogeological and groundwater assessment reports**.
- Coordinated field investigations and laboratory testing programs.

Key Projects

- Groundwater Control Study – Mashref-2 Compound, KSA
- Groundwater Control – AL Khor City, Qatar
- Groundwater Lowering Study – Mashref-1 Compound, KSA
- Groundwater Extraction – Kigali, Rwanda
- Groundwater Dewatering – Ismailia & Alexandria, Egypt
- Pumping Test Analysis – Ain Sokhna Port, Egypt
- Groundwater Hazards In Swelling Soil - New Capital, Egypt.

Project Manager of Groundwater Projects

United Group for Engineering Solution Company

2014 – 2017 | 2023 – Present (Per Task)

Key Achievements & Responsibilities

- Led development of **groundwater monitoring and instrumentation systems**.
- Managed drilling works, field investigations, and monitoring installations.
- Conducted **flash flood assessments and groundwater surveys**.
- Supervised pumping tests, tracer tests, and data interpretation.
- Prepared **method statements, technical documentation, and supervision reports**.

Key Projects

- Extraction Wells Survey – El Dabaa Nuclear Power Plant (NPPA), Egypt
- Piezometers & Monitoring Wells – NPPA, SCZONE & Ain Sokhna Port, Egypt
- Extraction and Recharge Wells – IDCO MDF Factory, Egypt
- Remediation Monitoring System – Abo Zabaal, Egypt
- Soakaway Design – Ain Sokhna Port, Egypt

- Soil Improvement Monitoring – Port Said Industrial Area, Egypt
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Hydrologist – Freelance

Manar Al-Maamar Engineering Consulting Company

2025

Key Achievements & Responsibilities

- Managed and delivered **flood protection projects** affiliated with **the Saudi Ministry of Interior**.
- Successfully obtained 13 official **approvals** from **10 different municipalities** across Saudi Arabia

Key Projects

- Flood Protection Project for Support Services Buildings- Madinah, Eastern Province, Tabuk, Algouf, KSA
 - Flood Protection Project for General Investigation Sites -Al-Khobar, Jazan, Assir, Baha, Najran, KSA
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Hydrogeologist – Freelance

Consultant Engineering Center

2021

Key Achievements & Responsibilities

- Prepared **hydrogeological and groundwater control study report**.

Key Projects

- Subsurface Water Investigation – Assiut Hydro Cracking Complex (AHC), Egypt
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Hydrogeologist – Freelance

Heliopolis University for Sustainable Development

2023

Key Achievements & Responsibilities

- Built and calibrated **groundwater models** to assess **climate change** and **pumping impacts** on large aquifer systems in Egypt.
- Evaluated **production and rejection wells** to support **sustainable groundwater management**.

Key Projects

- Groundwater Management Study – 2,500+ production wells, North East Sinai, Egypt
 - Groundwater Management Study – 622 production + 138 rejection wells, North East Sinai, Egypt
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Design Engineer – Concrete & Infrastructure

Soliman Consult – Engineering Office

2010 – 2012

Key Achievements & Responsibilities

- Designed and supervised **reinforced concrete and infrastructure works**.
- Prepared structural drawings, calculations, and technical reports.
- Coordinated designs with architectural, mechanical, and electrical disciplines.
- Conducted site supervision and resolved construction issues.

Key Projects

- Structural Design & Supervision – Gas Stations (Multiple Locations)
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Junior Civil Engineer

2009 – 2010

Key Achievements & Responsibilities

- Supported engineering design, quantity take-offs, and technical documentation.
- Assisted in site supervision and quality control activities.
- Coordinated between site execution teams and design offices.

Key Projects

- Building Construction & Infrastructure Support Projects

EDUCATION

Doctor of Philosophy (PhD) in Civil Engineering

Al-Azhar University, Cairo, Egypt

Year: 2025

Dissertation Title:

Modeling Groundwater Flow in Karst Aquifers and Interaction with Sedimentary Aquifers

Dissertation Focus

- **Groundwater flow modeling** in karst and sedimentary aquifer systems
- **Hydraulic connectivity** and aquifer interaction analysis
- Application of **MODFLOW** and **CFP** for complex hydrogeological systems in **karst aquifers**

Supervisors

- **Prof. Ahmed Hasan Imam**
- **Prof. Mostafa Ali**
- **Dr. Mohamed Attia**

Master of Science (MSc) in Civil Engineering

Al-Azhar University, Cairo, Egypt

Year: 2017

Thesis Title:

Analysis of Steady-State and Transient Recharge Inflow into Fractured Lithology

Thesis Description

- Integration of **hydrological modeling** and **groundwater modeling**
- Estimation of **direct rainfall recharge**
- Simulation of **stratigraphy**, **solute transport**, and **variable density flow**
- Application of **MODFLOW**, **MT3DMS**, and **SEAWAT**

Supervisors

- **Prof. Anas Elmolla**
- **Prof. Mohamed Farouk**
- **Dr. Walid Alsayed**
- **Dr. Rani Fouad**

Bachelor of Science (BSc) in Civil Engineering

Al-Azhar University, Cairo, Egypt

Year: 2009

- **Excellent with Honor (87.75%)**
- **Graduation Project: Design of Irrigation Works** (Excellent – 95%)

RESEARCH & PUBLICATIONS

- **Estimation of Runoff and Recharge Rates by Using Channel Routing Mechanism**
- **Experimental Comparison of Single and Dual Conduit Connectivity in Karst Aquifers**
- **Experimental Investigations of the Effect of Conduit Connectivity on Flow Exchange in Karst Aquifers**

PROFESSIONAL & RESEARCH EXPERIENCE (ACADEMIC CONTEXT)

Senior Hydrologist / Hydrogeologist (Applied Research & Consultancy)

- Conducted **hydrological and hydrogeological studies** supporting **academic research validation**.
- Applied **groundwater modeling** results to real-world case studies.
- Integrated **field investigations**, **pumping tests**, and **monitoring data** into research analysis.
- Contributed to **applied research** in **flood protection**, **groundwater control**, and **urban water systems**.

CERTIFICATIONS & COURSES

- **Groundwater Modeling System (GMS)**
- **SEAWAT**
- **MODPATH**
- **Arc Hydro Tools**
- **HYFRAN**
- **WaterGEMS**
- **Civil 3D**
- **SAP2000**
- **ETABS**
- **MODFLOW**
- **MT3DMS**
- **GIS**
- **HEC-HMS**
- **WMS**
- **SewerGEMS**
- **AutoCAD 2D**
- **SAFE**

SOFT SKILLS

- Academic Communication & Scientific Writing
- Research Design and Methodology Development
- Graduate and Undergraduate Student Mentoring
- Lecture Preparation & Knowledge Transfer
- Interdisciplinary Research Collaboration
- Time Management for Research & Teaching
- Professional Ethics & Academic Integrity
- Conference & Seminar Participation

LANGUAGES

- Arabic – Native
 - English – Professional Working Proficiency
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